

Classroom Observation Guide

Grade 6 Mathematics — Day Nine: "Area of a Parallelogram — Refine, Practice & Quiz"

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Prepared for visiting guests, administrators, and instructional observers.

Purpose of This Guide

Thank you for observing our Grade 6 mathematics classroom today. Day Eight proved the big idea that a leaning parallelogram is a rectangle in disguise — cut a triangle off one end, slide it to the other, and $\text{Area} = \text{base} \times \text{height}$ falls out of a shape students can watch being rebuilt. Day Nine does not introduce a new idea. It is the i-Ready **Refine** session: the class finishes Session 2's Apply It problems, works through Session 3's Refine example and guided practice (including a worked mistake from a student named Juanita), completes an Additional Practice page, and sits the full five-problem Lesson 1 Quiz before closing with a Math Journal entry. This guide explains how today's lesson was built, maps it directly back to the i-Ready Teacher Edition pages it is drawn from, and shows the standards it addresses.

The Design Decision: Refine Before Redirect

Day Eight's exit ticket and Enrich responses set today's opening focus: has the class internalized that the perpendicular height — never the slanted side — is the number that multiplies the base? Day Nine is built to answer that question with evidence, not a guess. Rather than moving the whole class to a brand-new shape before that question is settled, today keeps every student on parallelograms for one more period, but shifts the cognitive demand from *deriving* the formula (Day Eight's work) to *applying* it under increasing distraction: mixed-number bases, a worked-out student mistake to diagnose, a shape measured in tiles instead of inches, and a quiz problem that asks students to find **two** different correct heights for the very same parallelogram. Each of these is a distractor the i-Ready Teacher Edition names explicitly, and each one appears in today's app numbered exactly as it appears in the worktext.

From Proving the Formula to Trusting It Under Pressure

Day Eight asked, "why does $\text{Area} = \text{base} \times \text{height}$ work for a parallelogram?" and answered it with a cut-and-slide proof. Day Nine asks a harder, quieter question: "can a student still find the right base and height when the numbers get messier and a wrong answer is sitting right there, already worked out by someone else?" Refine, Problem 3 puts that question in front of students directly — Juanita's barn-window answer of 120 in.^2 is real, plausible, and wrong, because she multiplied the base by the slanted side instead of the height. Diagnosing her mistake, rather than only avoiding it themselves, is the version of precision (MP.6) this lesson is built around.

i-Ready Teacher Edition Alignment

Every problem in today's app is drawn directly from the i-Ready Classroom Mathematics Grade 6 Teacher Edition, Lesson 1 ("Find the Area of a Parallelogram"), continuing exactly where Day Eight's Explore → Model It → Try It portion of Session 2 left off. The table below maps the ten i-Ready Teacher Edition pages/sections used to build today's lesson onto the app's nine roadmap steps, in the order the class moves through them, so an observer can confirm page-for-page and number-for-number coverage of today's assigned material.

#	i-Ready Teacher Edition Content	Where It Appears in Today's App	Standard(s)
1	Session 2 • Apply It — Problems 6–8: the spinach-pie parallelogram, a select-all dimensions problem, and a grid parallelogram.	Step 2 — <i>Apply It: Finish Session 2</i>	6.G.A.1
2	Session 3 • Refine — Study the Example: a fully worked mixed-number problem (base $9\frac{1}{4}$ ft × height 4 ft = 37 ft ²).	Step 3 — <i>Refine: Worked Example</i> (worked example)	6.G.A.1
3	Session 3 • Refine — Guided Practice: a second mixed-number parallelogram for students to solve using the same method.	Step 3 — <i>Refine: Worked Example</i> ("Now You Try")	6.G.A.1, MP.2
4	Session 3 • Refine, Problem 3 & Error Alert — the barn-window problem and Juanita's base-times-slant mistake, named in the Teacher Edition by name.	Step 4 — <i>Refine: Juanita's Mistake</i>	6.G.A.1, MP.6
5	Additional Practice, Problem 3 — the tiled-wall mirror, measured in 4-inch square tiles instead of inches.	Step 5 — <i>Additional Practice 3–5</i> (Problem 3)	6.G.A.1, 3.MD.C.7d
6	Additional Practice, Problem 4 — "Which Side Is the Base?", a written-explanation item with no perpendicular tick given in the prompt.	Step 5 — <i>Additional Practice 3–5</i> (Problem 4)	6.G.A.1, MP.4
7	Additional Practice, Problem 5 — the Eagle Express Shipping Service logo, a real-world parallelogram with a slant-side distractor.	Step 5 — <i>Additional Practice 3–5</i> (Problem 5)	6.G.A.1
8	Lesson 1 Quiz, Problems 1–3 — grid-entry area with a diagonal distractor, the kitchen-tile problem, and a mixed-number area (6 $\frac{1}{5}$ ft).	Step 7 — <i>Lesson 1 Quiz (Problems 1–3)</i>	6.G.A.1
9	Lesson 1 Quiz, Problems 4–5 — Sayali's four-parallelogram sidewalk-chalk design, and the two-bases-same-parallelogram question.	Step 8 — <i>Lesson 1 Quiz (Problems 4–5)</i>	6.G.A.1, MP.2, MP.5
10	Math Journal, Error Alert & End of Lesson Checklist — the closing pages of Session 3 that ask students to draw their own area-12 parallelogram and self-assess against the lesson's "I Can" statements.	Step 9 — <i>Math Journal & Close</i>	6.G.A.1, MP.4, MP.6

Reading this table as an observer: the app's sidebar roadmap (visible throughout today's period) is numbered 1–9 and always shown to the whole class as the lesson is projected. Steps 2 through 9 above account for all ten i-Ready Teacher Edition pages assigned for today; Step 1, "Welcome Back & Warm-Up," is a two-minute bridge back to Day Eight and is not tied to a numbered worktext page. Every number, distractor, and correct answer named in the right-hand column above is taken directly from the Teacher Edition, so this guide — and the app itself — can be checked line-for-line against the physical worktext in hand.

Lesson at a Glance (55 Minutes)

Time	Segment	What's Happening
0–3 min	Welcome Back	Students sign in on the lesson app. A short bridge names today as Refine & Quiz day — same shape, same formula, harder distractors.
3–8 min	Apply It: Finish Session 2	Problems 6–8 close out Session 2: the spinach-pie parallelogram, a select-all dimensions problem, and a grid parallelogram.
8–16 min	Refine: Worked Example	The Refine session opens with a fully worked mixed-number example ($9\frac{3}{4}$ ft $\times$ 4 ft), then a matched practice problem with a new mixed-number base.
16–24 min	Refine: Juanita's Mistake	The barn-window problem and the Teacher Edition's own Error Alert: Juanita multiplied the base by the slanted side (10 in.) instead of the height, landing on 120 in. ² instead of the correct 72 in. ² .
24–33 min	Additional Practice 3–5	The tiled-wall mirror (tiles $\rightarrow$ inches), a written explanation of which side is the base, and the Eagle Express logo.
33–38 min	Engage / Explore / Enrich	Three boards grouped by how confidently each student separated the height from the slant during Refine and Additional Practice.
38–44 min	Lesson 1 Quiz (1–3)	Grid-entry area with a diagonal distractor, the kitchen-tile problem, and a mixed-number area — three of the quiz's five problems.
44–50 min	Lesson 1 Quiz (4–5)	Sayali's four-parallelogram chalk design (find the unshaded area) and the two-bases-same-parallelogram question, worth two points each.
50–55 min	Math Journal & Close	Students draw and label their own area-12 cm ² parallelogram, then self-assess against five "I Can" checklist statements.

Why Refine With a Named Mistake, Not Just More Problems?

It would be simple to hand students ten more parallelogram problems and call it practice. Instead, the Teacher Edition — and today's app, following it exactly — puts a specific wrong answer in front of students and asks them to find where it went wrong before revealing the explanation. Diagnosing someone else's mistake requires a student to hold the correct method in mind well enough to compare it against a plausible alternative, which is a meaningfully harder task than solving a fresh problem correctly on the first try. This is the clearest way to surface whether the height-vs-slant distinction from Day Eight actually transferred, rather than assuming it did because students appeared to follow along.

Today's Classwork Worksheet: "Two Sides, Two Heights, Same Area"

P E R _ E N D I C U _ A R → PERPENDICULAR

Each student receives a one-page classwork worksheet (CCSS 6.G.A.1, 3.MD.C.7d, L.6.4, MP.2, MP.4, MP.5, MP.6). Section A, Guess the Word, has students fill in two missing letters to spell PERPENDICULAR — the exact term the app and worktext use for a parallelogram's true height, continuing the word-then-model design used for DECOMPOSE on Day Seven and REARRANGE on Day Eight. Section B, word bank sentences, uses *parallelogram*, *base*, *height*, and *perpendicular* to define the shape and its parts in the exact terms the app uses. Section C walks students through a worked example (sides 6 and 4, two valid base/height pairs, area 24) and then asks them to solve their own parallelogram problem (sides 5 and 6, area 30) — deliberately different numbers from Quiz Problem 5's 36-and-4 parallelogram, so the worksheet teaches the "two honest base/height pairs" idea without giving away the quiz answer. A closing Refine · Summary bar previews next: cutting a parallelogram in half along a diagonal produces two triangles, each exactly half the area — the on-ramp into the unit's next topic, Area of Triangles.

The Engage / Explore / Enrich Boards

Board	Who	Task	IXL Skill
Engage (with teacher)	Still reaching for the slant or a diagonal instead of the true height.	Cut a 12-by-5 paper rectangle along a slanted segment and slide the triangle to confirm the resulting parallelogram still has base 12, height 5, area 60; redo the barn window together and circle the right-angle mark before multiplying.	In-class only — no IXL assignment; a physical, teacher-guided repair of the height-vs-slant misconception.
Explore (independent)	Comfortable with whole-number base × height, ready for mixed numbers and distractors.	Additional Practice, Problems 3–5; Lesson 1 Quiz, Problems 1–3.	GG.4 – Area of parallelograms
Enrich (independent, finished early)	Fluent with base × height, ready for Quiz Problem 5's twist.	A parallelogram with area 24 sq units and sides 36 and 4 units: find BOTH possible heights and explain why both describe the same parallelogram; full Lesson 1 Quiz, Problems 1–5.	GG.4 – Area of parallelograms

IXL codes are from IXL's published Grade 6 skill list under Perimeter and Area (GG).

Common Core & Michigan Standards Addressed

Michigan's K–12 Standards for Mathematics adopt the Common Core State Standards; the codes below are identical in both documents.

Grade-Level Standard (Grade 6)

6.G.A.1: Find the area of right triangles, other triangles, special quadrilaterals, and polygons by composing into rectangles or decomposing into triangles and other shapes; apply these techniques to real-world problems. Today's ten problems — from the spinach pie through the Lesson 1 Quiz — are this standard applied to the parallelogram case across increasing levels of numeric and contextual difficulty.

Foundational Standard Being Extended

3.MD.C.7d: Recognize area as additive; find the area of rectilinear figures by decomposing them into non-overlapping rectangles and adding the areas of the non-overlapping parts, applying this technique to solve real-world problems. The tiled-wall mirror (Additional Practice, Problem 3) and Sayali's four-parallelogram design (Quiz, Problem 4) both require students to work with area as additive quantities built from repeated or combined parallelogram regions.

ELA/Literacy Standard Addressed

L.6.4: Determine or clarify the meaning of unknown and multiple-meaning words and phrases, choosing flexibly from a range of strategies. Section A of the classwork worksheet has students infer PERPENDICULAR from its letter pattern and context before applying it to today's task.

Standards for Mathematical Practice

MP.2 (Reason abstractly and quantitatively): Splitting a mixed-number base like $9\frac{1}{4}$ into $9 + \frac{1}{4}$ and multiplying each part by the height, then reconnecting that number back to the shape's actual area, asks students to move fluently between the symbolic and the geometric.

MP.4 (Model with mathematics): Additional Practice, Problem 4 asks students to explain in writing which side is the base and why — not just compute an answer — requiring them to put their geometric reasoning into a model other people can follow.

MP.5 (Use appropriate tools strategically): Quiz Problem 5 is solved by dividing the given area by each of the two side lengths in turn — choosing division as the tool that undoes multiplication, applied twice, to find both valid heights of the same parallelogram.

MP.6 (Attend to precision): Every problem today places a real, true measurement — a slanted side, a diagonal, an “other side” length — next to the base and height and asks students to use only the correct two. Diagnosing Juanita's mistake is this standard in its sharpest form: precision isn't just computing correctly, it's catching an error that used the wrong number correctly.

Pedagogical & Cognitive Research Foundations

Refine Sessions and Distributed, Effortful Retrieval

Roediger and Karpicke (2006) found that retrieval practice — actively recalling and applying a procedure rather than re-reading it — produces more durable learning than repeated exposure alone. The i-Ready Refine session is built on exactly this principle: rather than re-teaching the cut-and-slide proof, today's Refine example and guided problem ask students to retrieve and reapply base $\times$ height under new conditions (a mixed-number base) the very next day.

Learning From a Worked Error

Durkin and Rittle-Johnson (2012) found that having students compare a correct and an incorrect worked example — explaining why the incorrect one fails — produces stronger conceptual understanding than studying correct examples alone. Juanita's barn-window mistake gives today's lesson exactly this structure: students first solve the problem correctly, then are shown Juanita's answer and asked to reconstruct her reasoning before the Teacher Edition's Error Alert confirms it.

The Height-vs-Slant Misconception Is a Spatial Reasoning Issue

Battista (2007) documents that students commonly rely on a shape's visually salient boundary — here, the long slanted side — rather than a geometrically defined measurement like perpendicular height. Today's problems each label the slant, an “other side,” or a diagonal explicitly as a value that is real but unused, giving repeated, varied practice at making that distinction under different visual dressings (a logo, a tiled wall, a chalk design).

Two Base/Height Pairs, One Parallelogram: Building Flexibility

Rittle-Johnson and Star (2007) found that comparing multiple correct solution paths on the same problem builds stronger flexible understanding than studying one path in isolation. Quiz Problem 5 applies this directly to geometry: the same 24-square-unit parallelogram has two entirely different, equally correct base/height pairs, and finding both is what the problem actually assesses.

What to Look For as an Observer

- Students using the words “base,” “height,” and “perpendicular” — not “the long side” or “the slanted part” — when explaining their work.
- During Refine, students attempting to reconstruct Juanita's mistake before the Error Alert is revealed, rather than jumping straight to the correct answer.
- During Additional Practice, Problem 4, whether students can write a clear justification for which side is the base, not just circle one.
- At the tiled-wall mirror problem, students remembering to convert tiles to inches before multiplying, not after.
- On Quiz Problem 5, whether a student attempts only one base/height pair or checks both before answering — this is the single clearest formative signal from today's quiz.
- Which students still need the slant distractor pointed out versus which identify it independently — this sets tonight's Engage/Explore/Enrich placement and Day Ten's opening observer note.

Appendix: Day Nine in the Grade 6 Math Lesson Plan Template

Template Field	Day Nine Content
Standard(s) & Taxonomy Level	6.G.A.1 (grade level); 3.MD.C.7d (foundational); L.6.4 (ELA/literacy, classwork worksheet); MP.2, MP.4, MP.5, MP.6. Apply / Analyze (applying the parallelogram formula under distraction and diagnosing an incorrect worked solution).
Learning Target(s) (SWBAT)	SWBAT find the area of a parallelogram given a mixed-number base or height; SWBAT identify the specific error in an incorrect worked solution that used a side length instead of the height; SWBAT find both valid base/height pairs for a single parallelogram.
Success Criteria (I Can)	I can apply $\text{Area} = \text{base} \times \text{height}$ to mixed numbers and multi-step problems. I can catch a base-times-slant mistake like Juanita's before I make it. I can see that a parallelogram has two honest base/height pairs. I can explain in writing which side is the base and why.
Daily Marzano Strategies	Identifying Critical Content; Examining Reasoning (diagnosing Juanita's error); Practicing Skills, Strategies, and Processes; Cooperative Learning (Engage/Explore/Enrich rotation).
Do-Now (5 min)	Welcome Back bridge restating Day Eight's proof and naming today's two named watch-fors: the slant-vs-height mistake and the two-bases idea.
Instructional Plan	Apply It (finish Session 2) → Refine worked example & guided practice → Refine Problem 3 & Error Alert (Juanita) → Additional Practice 3–5 → Engage/Explore/Enrich → Lesson 1 Quiz (all 5 problems) → Math Journal & Checklist.
Critical Content Debrief (5 min)	Can students state, in their own words, exactly what Juanita did wrong? Can they find both correct heights for the Quiz Problem 5 parallelogram, not just one?
Daily Formative [Ungraded] / Summative [Graded] Assessments	Formative: teacher observation during Refine and Additional Practice, classwork worksheet. Summative: the full five-problem Lesson 1 Quiz, graded, closing this lesson's content.

Planning to Close the Achievement Gap Using Data

Tier 1 Supports	Tier 2 Supports	Tier 3 Supports
Whole-class Refine example walkthrough on the document camera before independent work; every diagram labels the slant, diagonal, or “other side” as a distractor; the classwork worksheet’s word bank is letter-boxed for every student.	Engage board with the teacher: physically cut and slide a paper parallelogram; measure height with a set square against the base; sentence frames naming base, height, and slant before attempting Juanita’s problem independently.	Support Room: Multiplication & Division Fluency (continued from Day Seven, run alongside — not instead of — today’s full grade-6 Refine and Quiz content). Placement is revisited weekly against new formative data.

Data Source: Day Eight Model It, Practice, and Exit Ticket results; Day Eight Observation Guide recommendation; Day Nine Refine, Additional Practice, and Lesson 1 Quiz results.

Continuity Note

This guide carries forward the recommendation made after observing Day Eight and applies it to today’s Refine-and-Quiz design. Day Nine’s own formative results — especially which students diagnose Juanita’s mistake independently and which find both base/height pairs on Quiz Problem 5 — will in turn shape the observer note that opens Day Ten, where the same cut-and-rearrange reasoning extends from parallelograms to triangles.

Explore More: Resources for Observers

IXL Skill Alignment, Grade 6 The published Grade 6 skill list under Perimeter and Area (GG), used to choose today’s Engage/Explore/Enrich board tasks. ixl.com/math/grade-6	Michigan K–12 Math Standards The Michigan Department of Education’s full K–12 Standards for Mathematics document. View Michigan DOE standards
Today’s Classwork Worksheet “Two Sides, Two Heights, Same Area,” the Guess the Word / word bank worksheet referenced throughout this guide. View the worksheet	Standards for Mathematical Practice The official Common Core description of MP.2, MP.4, MP.5, MP.6, and the other four practices. View on corestandards.org

References

Battista, M. T. (2007). The development of geometric and spatial thinking. In F. K. Lester Jr. (Ed.), *Second handbook of research on mathematics teaching and learning* (pp. 843–908). Information Age Publishing.

Durkin, K., & Rittle-Johnson, B. (2012). The effectiveness of using incorrect examples to support learning about decimal magnitude. *Learning and Instruction*, 22(3), 206–214.

National Governors Association Center for Best Practices & Council of Chief State School Officers. (2010). *Common Core State Standards for Mathematics*.

Rittle-Johnson, B., & Star, J. R. (2007). Does comparing solution methods facilitate conceptual and procedural knowledge? *Journal of Educational Psychology*, 99(3), 561–574.

Roediger, H. L., & Karpicke, J. D. (2006). Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, 17(3), 249–255.

For the response-to-intervention research base supporting the continued multiplication/division support-room block, see the References section of the Day Seven Observation Guide.